

Open Problem Note:

Exact Full Indifferentiability of 6-Round Feistel Networks

Theoretical Cryptography Note

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Abstract

The Feistel construction is a standard mechanism for converting public random functions into public pseudorandom permutations. While 5-round Feistel is provably non-indifferentiable from a random permutation (Coron, Patarin & Seurin, CRYPTO 2008), and 8-round Feistel is fully indifferentiable (Dai & Steinberger, CRYPTO 2016), the exact full indifferentiability status of 6-round Feistel remains one of the longest-standing open problems in symmetric-key theoretical cryptography.

1 Formal Setup

Let $\mathcal{F}_n = \{F : \{0, 1\}^n \rightarrow \{0, 1\}^n\}$ denote the set of all n -bit to n -bit functions, and let $\mathcal{P}_{2n} = \text{Perm}(\{0, 1\}^{2n})$ denote the set of all $2n$ -bit permutations.

Definition 1 (Feistel Network). *Given r round functions $\mathbf{F} = (F_1, \dots, F_r) \in (\mathcal{F}_n)^r$, the r -round Feistel construction $\Psi^r[\mathbf{F}] : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{2n}$ maps $(L_0, R_0) \mapsto (L_r, R_r)$ via the iterative update rule for $i = 1, \dots, r$:*

$$L_i = R_{i-1}, \quad R_i = L_{i-1} \oplus F_i(R_{i-1}). \quad (1)$$

Definition 2 (Indifferentiability Framework (Maurer, Renner & Holenstein, TCC 2004)). *A construction $C^{\mathbf{F}}$ using oracle access to ideal components $\mathbf{F}$ is **indifferentiable** from an ideal primitive $\mathcal{P}$ if there exists a stateful PPT simulator $\mathcal{S}$ with oracle access to $\mathcal{P}$ such that for every PPT distinguisher $\mathcal{D}$:*

$$\text{Adv}_{C, \mathcal{S}}^{\text{indiff}}(\mathcal{D}) := \left| \Pr \left[\mathcal{D}^{C^{\mathbf{F}}, \mathbf{F}}(1^n) = 1 \right] - \Pr \left[\mathcal{D}^{\mathcal{P}, \mathcal{S}^{\mathcal{P}}}(1^n) = 1 \right] \right| \leq \epsilon(n), \quad (2)$$

where $\epsilon(n)$ is negligible in the security parameter n .

2 The 6-Round Open Conjecture

Conjecture 1 (Full Indifferentiability of 6-Round Feistel). *Let $\mathbf{F} = (F_1, \dots, F_6) \stackrel{\$}{\leftarrow} (\mathcal{F}_n)^6$ be independent random functions, and let $\mathcal{P} \stackrel{\$}{\leftarrow} \mathcal{P}_{2n}$ be a uniform random permutation.*

There exists an efficient simulator $\mathcal{S}^{\mathcal{P}}$ making at most $\text{poly}(q)$ queries to $\mathcal{P}$ such that for any distinguisher $\mathcal{D}$ making at most q total queries to its oracles:

$$\text{Adv}_{\Psi^6, \mathcal{S}}^{\text{indiff}}(\mathcal{D}) \leq O \left(\frac{q^k}{2^n} \right), \quad (3)$$

for some universal polynomial degree $k \geq 2$.

Rounds (r)	Status	Security Bound	Reference
$r \leq 4$	Attackable	Non-Indifferentiable	folklore, <i>a fortiori</i> from the $r = 5$ attack below
$r = 5$	Attackable	Non-Indifferentiable	Coron, Patarin & Seurin (2008)
$r = 6$	OPEN	Public / Sequential Indiff. Only	Mandal, Patarin & Seurin (2012)
$r \geq 8$	Proven	$O(q^8/2^n)$	Dai & Steinberger (2016)

Table 1: Indifferentiability status of r -round Feistel networks.

3 Current Bounds and Theoretical Barrier

Key Open Question: Can a simulator construct consistent responses for queries $F_1, \dots, F_6$ without incurring exponential blowup in query history when handling middle-round path completions (F_3, F_4) , or does a non-trivial distinguishing attack exist for $r = 6$?